

# AbangCebu AI “ Sprint 2 Master Plan & Execution Specification (Granular Module Breakdown) **Document Version:** 2.0.0 **Status:** Approved for Sprint 2 Execution **Project:** AbangCebu AI (`abang-cebu-ai`) **Target Sprint:** Sprint 2 (Sprint ID: `70` / Board ID: `1`) **Architecture Granularity:** Option B (Granular Feature/Screen-Level Modules) **Author:** Hermar Centillas (Lead / Scrum Master) **Co-Authors:** Engineering Team & QA Lead (Jenny Villamor) **Created Date:** October 1, 2026 --- ## 1. Executive Summary & Granularity Rationale In response to academic and project review requirements specifying a **strict per-module breakdown**, the system architecture is decomposed into **16 granular, self-contained functional modules** (Option B). Under this model: - Sub-processes previously grouped under parent umbrellas (such as Login, Registration, and Password Reset) are formalized as **independent system modules** with their own workflows, actors, database interactions, error states, and dedicated `.drawio` flowchart specifications. - Features with dual interfaces (such as Interactive Map Discovery vs Structured Attribute Filter Search) are cleanly decoupled to provide modular clarity for developers, testers, and academic evaluators. --- ## 2. Section A: Final Granular Module List (16 Modules) | # | Module Name | System Description & Scope | Current Status | Primary Evidence / Source |

|   |                                  |                                                                                                                                                                                                                   |                                 |                                                                                                                                                                                                                                                                                                                                                                                                      |
|---|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | User Registration Module         | Renter and Landlord account creation, role selection, NIST SP 800-63B password complexity, Philippine mobile format normalization (+639XXXXXXXX), Turnstile bot validation, and PKCE verification email dispatch. | Foundation Implemented / Pre-UI | [`src/types/auth.ts`](file:///home/hrmr/abang-cebu-ai/src/types/auth.ts), [`docs/specifications/auth/auth-registration-spec.md`](file:///home/hrmr/abang-cebu-ai/docs/specifications/auth/auth-registration-spec.md), [`docs/design/registration-page-wireframe.md`](file:///home/hrmr/abang-cebu-ai/docs/design/registration-page-wireframe.md)                                                     |
| 2 | User Login & Session Module      | User authentication, Supabase GoTrue JWT exchange, HTTP-only cookie chunking (sb-auth-token), Refresh Token Rotation (RTR), multi-tab synchronization (BroadcastChannel), and role-based route redirects.         | Foundation Implemented / Pre-UI | [`src/types/auth.ts`](file:///home/hrmr/abang-cebu-ai/src/types/auth.ts), [`middleware.ts`](file:///home/hrmr/abang-cebu-ai/middleware.ts), [`docs/specifications/auth/auth-session-lifecycle.md`](file:///home/hrmr/abang-cebu-ai/docs/specifications/auth/auth-session-lifecycle.md), [`docs/design/login-page-wireframe.md`](file:///home/hrmr/abang-cebu-ai/docs/design/login-page-wireframe.md) |
| 3 | Password Reset & Recovery Module | Self-service password recovery, anti-enumeration security timing, single-use PKCE recovery email dispatch, password updates, and mandatory global revocation of older sessions (scope: 'global').                 | Foundation Implemented / Pre-UI | [`docs/specifications/auth/auth-password-reset-spec.md`](file:///home/hrmr/abang-cebu-ai/docs/specifications/auth/auth-password-reset-spec.md), [`docs/testing/auth-test-plan.md`](file:///home/hrmr/abang-cebu-ai/docs/testing/auth-test-plan.md)                                                                                                                                                   |
| 4 | User Profile Management Module   | Management of `public.profiles`, personal information updates (full name, phone number, bio), avatar image                                                                                                        |                                 |                                                                                                                                                                                                                                                                                                                                                                                                      |

uploads to Supabase storage with MIME/size limits, and account preferences. |  
\*\*Foundation Implemented / In Development\*\* | [docs/database/users-and-profiles-schema.md](file:///home/hrmr/abang-cebu-ai/docs/database/users-and-profiles-schema.md), [src/types/database.ts](file:///home/hrmr/abang-cebu-ai/src/types/database.ts), [docs/design/profile-page-wireframe.md](file:///home/hrmr/abang-cebu-ai/docs/design/profile-page-wireframe.md) | | \*\*5\*\* | \*\*Property Listing Creation Module\*\* | Landlord property creation wizard: property metadata (building type, address, barangay, city, PostGIS coordinates), multi-unit configurations (rent, deposit, rooms, rules), amenity tagging, and photo uploads. |  
\*\*Schema Defined / Planned\*\* | [docs/database/database-erd.md](file:///home/hrmr/abang-cebu-ai/docs/database/database-erd.md) (`properties`, `rental\_units`, `property\_photos`, `amenities`), [docs/architecture/what-is-abangcebu-ai.md](file:///home/hrmr/abang-cebu-ai/docs/architecture/what-is-abangcebu-ai.md) | |  
\*\*6\*\* | \*\*Unit Availability & Management Module\*\* | Landlord listing dashboard: toggling real-time room/bedspace occupancy (`is\_available`), adjusting monthly rental rates, archiving units, and reviewing unit-specific views and inquiries. |  
\*\*Schema Defined / Planned\*\* | [docs/database/database-erd.md](file:///home/hrmr/abang-cebu-ai/docs/database/database-erd.md) (`rental\_units.is\_available`), [docs/design/user-dashboard-wireframe.md](file:///home/hrmr/abang-cebu-ai/docs/design/user-dashboard-wireframe.md) | | \*\*7\*\* | \*\*Interactive Map Discovery Module\*\* | Geospatial rental visualization across Metro Cebu using MapLibre GL: viewport bounding-box synchronization (`&& ST\_MakeEnvelope`), interactive price pins, geolocation centering, and pin selection preview drawers. |  
\*\*Schema Defined / Planned\*\* | [docs/database/database-erd.md](file:///home/hrmr/abang-cebu-ai/docs/database/database-erd.md) (PostGIS section), [docs/design/mobile-map-wireframes.md](file:///home/hrmr/abang-cebu-ai/docs/design/mobile-map-wireframes.md), [src/components/map/](file:///home/hrmr/abang-cebu-ai/src/components/map/) | | \*\*8\*\* | \*\*Property Search & Filter Module\*\* | Parametric multi-facet search engine: filtering by budget min/max, room category (bedspace, room, condo), gender rules, curfew, visitor policy, utilities, and landmark radius (`ST\_DWithin`). |  
\*\*Schema Defined / Planned\*\* | [docs/architecture/what-is-abangcebu-ai.md](file:///home/hrmr/abang-cebu-ai/docs/architecture/what-is-abangcebu-ai.md), [docs/database/database-erd.md](file:///home/hrmr/abang-cebu-ai/docs/database/database-erd.md) | | \*\*9\*\* | \*\*AI Rental Assistant Module\*\* | Natural language conversational finder supporting English and Cebuano/Bisaya; translates prompts into structured query filters; resolves ambiguous landmarks; strictly grounded in approved database listings. |  
\*\*Behavioral Spec Defined / Planned\*\* | [docs/architecture/what-is-abangcebu-ai.md](file:///home/hrmr/abang-cebu-ai/docs/architecture/what-is-abangcebu-ai.md) (Sections: "The Role of AI", "How It Works") | | \*\*10\*\* | \*\*Inquiry & Viewing Reservation Module\*\* | Manages renter inquiries on properties and schedules in-person ocular viewing visits (`viewing\_date`); handles landlord acceptance/rescheduling state machine. Excludes financial payment processing. |  
\*\*Schema Defined / Planned\*\* | [docs/database/database-erd.md](file:///home/hrmr/abang-cebu-ai/docs/database/database-erd.md)

hrmr/abang-cebu-ai/docs/database/database-erd.md) (`inquiries` table), [docs/architecture/what-is-abangcebu-ai.md](file:///home/hrmr/abang-cebu-ai/docs/architecture/what-is-abangcebu-ai.md) | | **\*\*11\*\*** | **\*\*Landlord KYC & Verification Module\*\*** | Anti-scam trust layer: Landlord government ID upload (PhilSys, Passport, PRC, UMID) and proof of property ownership; private storage RLS; administrative review queue before public listing publishing. | **\*\*Schema & Types Defined / Planned\*\*** | [docs/database/database-erd.md](file:///home/hrmr/abang-cebu-ai/docs/database/database-erd.md) (`kyc\_verifications`), [src/types/database.ts](file:///home/hrmr/abang-cebu-ai/src/types/database.ts) (`KycStatus`, `KycIdType`) | | **\*\*12\*\*** | **\*\*Messaging & Contact Module\*\*** | Direct communication channel between renters and verified landlords; exposes verified phone numbers, direct contact links, and inquiry conversation threads. | **\*\*Architectural Scope Defined / Planned\*\*** | [docs/architecture/what-is-abangcebu-ai.md](file:///home/hrmr/abang-cebu-ai/docs/architecture/what-is-abangcebu-ai.md), [docs/database/database-erd.md](file:///home/hrmr/abang-cebu-ai/docs/database/database-erd.md) | | **\*\*13\*\*** | **\*\*Review & Rating Module\*\*** | Verified tenant evaluation workflow: 1-to-5 star ratings, qualitative feedback, verified tenant badges, and aggregate rating score calculations on properties. | **\*\*Schema Defined / Planned\*\*** | [docs/database/database-erd.md](file:///home/hrmr/abang-cebu-ai/docs/database/database-erd.md) (`reviews` table) | | **\*\*14\*\*** | **\*\*Reporting & Scam Flagging Module\*\*** | Community moderation tool: allows users to flag fraudulent listings, inaccurate pricing, or abusive landlords; triggers immediate administrative alerts in audit queue. | **\*\*Schema Defined / Planned\*\*** | [docs/database/database-erd.md](file:///home/hrmr/abang-cebu-ai/docs/database/database-erd.md) (`audit\_logs`), [docs/architecture/what-is-abangcebu-ai.md](file:///home/hrmr/abang-cebu-ai/docs/architecture/what-is-abangcebu-ai.md) | | **\*\*15\*\*** | **\*\*Admin Management & Moderation Module\*\*** | Central governance portal for administrators: listing approval/rejection queue, landlord KYC verification review, account suspensions (`is\_suspended`), and immutable audit log inspection. | **\*\*Wireframe & Schema Defined / Planned\*\*** | [docs/design/admin-dashboard-wireframe.md](file:///home/hrmr/abang-cebu-ai/docs/design/admin-dashboard-wireframe.md), [docs/database/database-erd.md](file:///home/hrmr/abang-cebu-ai/docs/database/database-erd.md) (`audit\_logs`), [src/types/database.ts](file:///home/hrmr/abang-cebu-ai/src/types/database.ts) (`is\_admin`) | | **\*\*16\*\*** | **\*\*Notification Module\*\*** | Transactional alerting system: dispatches email and in-app badge notifications for inquiry status transitions, viewing confirmations, KYC verification outcomes, and security alerts. | **\*\*Architectural Scope Defined / Planned\*\*** | [docs/specifications/auth/auth-registration-spec.md](file:///home/hrmr/abang-cebu-ai/docs/specifications/auth/auth-registration-spec.md), [docs/database/database-erd.md](file:///home/hrmr/abang-cebu-ai/docs/database/database-erd.md) | --- ## 3. Section B: Flowchart Deliverables Directory (docs/flowcharts/) Each of the 16 modules receives its own dedicated `.drawio` file, adopting standard uncompressed XML and clean monochrome formatting: ``text docs/â”€â”€â”€â”€ flowcharts/â”€â”€â”€â”€ user-registration.drawio # Module 1:

Registration, Validation & PKCE Activation â”œâ”€â”€ user-login.drawio # Module  
 2: Login, Session Token & Role Routing â”œâ”€â”€ password-reset.drawio #  
 Module 3: Self-Service Password Recovery & Invalidation â”œâ”€â”€ user-profile-  
 management.drawio # Module 4: Profile Details & Avatar Upload â”œâ”€â”€  
 property-listing-creation.drawio # Module 5: Landlord Multi-Unit Listing  
 Submission â”œâ”€â”€ unit-availability.drawio # Module 6: Real-Time Occupancy  
 & Unit Controls â”œâ”€â”€ interactive-map.drawio # Module 7: MapLibre  
 Viewport Panning & Spatial Query â”œâ”€â”€ property-search.drawio # Module 8:  
 Multi-Facet Filter & Radius Search â”œâ”€â”€ ai-rental-assistant.drawio # Module  
 9: Natural Language Parsing & Recommendations â”œâ”€â”€ inquiry-  
 reservation.drawio # Module 10: Ocular Viewing Requests & Scheduling State  
 â”œâ”€â”€ landlord-verification.drawio # Module 11: Government ID Submission  
 & Trust Pipeline â”œâ”€â”€ messaging-contact.drawio # Module 12: Direct Inquiry  
 Communication & Contact â”œâ”€â”€ review-rating.drawio # Module 13: Verified  
 Tenant Ratings & Reviews â”œâ”€â”€ report-moderation.drawio # Module 14:  
 Fraud Reporting & Moderation Flagging â”œâ”€â”€ admin-management.drawio #  
 Module 15: Admin Governance, Moderation & Audit Logs â””â”€â”€  
 notification.drawio # Module 16: Transactional Alerts & In-App Notices `` -- ## 4.  
 Section C: Sprint 2 Task Breakdown (Option B Alignment) ``text SPRINT 2 TASK  
 STRUCTURE (16 GRANULAR MODULES) â”œâ”€â”€ STREAM 1: Analysis &  
 Architecture Finalization â”œâ”€â”€ STREAM 2: Flowchart Authoring (.drawio  
 across 16 files) â”œâ”€â”€ STREAM 3: Feature Development (UI, Server Actions,  
 Route Handlers) â”œâ”€â”€ STREAM 4: Quality Assurance & Testing â””â”€â”€  
 STREAM 5: Review, Audit & Release Verification `` ### Stream 1: Analysis &  
 Architecture Finalization - \*\*TASK-S2-01: Module Architecture Finalization  
 (Option B)\*\* (0.5d / Lead) Finalize 16 granular modules in documentation; confirm  
 exclusion of payment processing. - \*\*TASK-S2-02: Supabase Schema Migration for  
 Marketplace Entities\*\* (1.5d / Senior Fullstack) Deploy SQL migration for  
 remaining tables: `properties`, `rental\_units`, `property\_photos`, `amenities`,  
 `property\_amenities`, `inquiries`, `reviews`, `saved\_listings`, `audit\_logs`. ###  
 Stream 2: Documentation & Flowchart Authoring - \*\*TASK-S2-03: Create `docs/  
 flowcharts/` & Author Authentication & User Flowcharts (4 Files)\*\* (1.5d / Lead)  
 Author `user-registration.drawio`, `user-login.drawio`, `password-reset.drawio`,  
 `user-profile-management.drawio`. - \*\*TASK-S2-04: Author Property & Search  
 Flowcharts (4 Files)\*\* (1.5d / Lead) Author `property-listing-creation.drawio`, `unit-  
 availability.drawio`, `interactive-map.drawio`, `property-search.drawio`. - \*\*TASK-  
 S2-05: Author AI, Inquiry & Trust Flowcharts (4 Files)\*\* (1.5d / Lead) Author `ai-  
 rental-assistant.drawio`, `inquiry-reservation.drawio`, `landlord-  
 verification.drawio`, `messaging-contact.drawio`. - \*\*TASK-S2-06: Author Review,  
 Moderation & Notification Flowcharts (4 Files)\*\* (1.5d / Lead) Author `review-  
 rating.drawio`, `report-moderation.drawio`, `admin-management.drawio`,  
 `notification.drawio`. ### Stream 3: Feature Development (Implementation) -  
 \*\*TASK-S2-07: Registration & Login UI Implementation\*\* (2.0d / Frontend  
 Engineer) Build interactive forms matching `registration-page-wireframe.md` and

`login-page-wireframe.md`. - **\*\*TASK-S2-08: User Profile & Avatar Storage Integration\*\*** (1.5d / Fullstack Engineer) Implement profile update Server Action and Supabase avatar upload bucket. - **\*\*TASK-S2-09: MapLibre GL Canvas & Viewport Bounds Integration\*\*** (2.5d / Fullstack Engineer) Connect MapLibre map to Supabase PostGIS `ST\_MakeEnvelope` query function. - **\*\*TASK-S2-10: Multi-Unit Property Listing Wizard\*\*** (2.0d / Fullstack Engineer) Implement landlord listing creation wizard and unit configuration forms. - **\*\*TASK-S2-11: Landlord KYC Document Upload & Admin Review Screen\*\*** (1.5d / Fullstack Engineer) Implement document upload with private bucket RLS and Admin approval actions. **### Stream 4: Quality Assurance & Testing** - **\*\*TASK-S2-12: Author Sprint 2 Master QA Test Plan\*\*** (1.5d / QA Lead) Model after `docs/testing/auth-test-plan.md` across all 16 granular modules. - **\*\*TASK-S2-13: Execute Automated & Manual Test Cases\*\*** (2.0d / QA Team) Run Vitest validation tests, PostGIS spatial queries, and Playwright integration tests. **### Stream 5: Review, Audit & Release Verification** - **\*\*TASK-S2-14: Security Audit of Storage & PostgreSQL RLS Policies\*\*** (1.0d / Security Auditor) Verify zero-trust policies for private KYC documents, listing access, and admin actions. - **\*\*TASK-S2-15: Release Packaging & Presentation Package\*\*** (0.5d / Lead) Package flowcharts, test execution matrices, and documentation for academic defense. --- **## 5. Section D: Master QA Test Plan (Option B Mappings)** All 16 granular modules are mapped to corresponding test suites following the structure in `docs/testing/auth-test-plan.md`: ``

| +-----+   GRANULAR QA TEST SUITE TRACEABILITY MATRIX   +-----+----- |                                  |  |                                                                       |
|---------------------------------------------------------------------|----------------------------------|--|-----------------------------------------------------------------------|
| +-----+   Module #                                                  | Granular Module Name             |  | Covered QA Suite & Test Case IDs   +-----+-----                       |
| +-----+   <b>**Module 01**</b>                                      | User Registration Module         |  | Suite 1: Registration (`TC-REG-01..07`)     <b>**Module 02**</b>      |
|                                                                     | User Login & Session Module      |  | Suite 2: Login & Session (`TC-LOG-01..08`)     <b>**Module 03**</b>   |
|                                                                     | Password Reset & Recovery Module |  | Suite 3: Recovery (`TC-RST-01..05`)     <b>**Module 04**</b>          |
|                                                                     | User Profile Management Module   |  | Suite 4: Profiles (`TC-PROF-01..04`)     <b>**Module 05**</b>         |
|                                                                     | Property Listing Creation Module |  | Suite 5: Listing Creation (`TC-PROP-01..04`)     <b>**Module 06**</b> |
|                                                                     | Unit Availability Module         |  | Suite 6: Unit Controls (`TC-AVAIL-01..03`)     <b>**Module 07**</b>   |
|                                                                     | Interactive Map Discovery Module |  | Suite 7: Map Spatial (`TC-MAP-01..04`)     <b>**Module 08**</b>       |
|                                                                     | Property Search & Filter Module  |  | Suite 8: Filters & Search (`TC-SRCH-01..04`)     <b>**Module 09**</b> |
|                                                                     | AI Rental Assistant Module       |  | Suite 9: AI Grounding (`TC-AI-01..04`)     <b>**Module 10**</b>       |
|                                                                     | Inquiry & Viewing Reservation    |  | Suite 10: Inquiries (`TC-INQ-01..04`)     <b>**Module 11**</b>        |
|                                                                     | Landlord KYC & Verification      |  | Suite 11: Trust & KYC (`TC-KYC-01..04`)     <b>**Module 12**</b>      |
|                                                                     | Messaging & Contact Module       |  | Suite 12: Communication (`TC-MSG-01..03`)     <b>**Module 13**</b>    |
|                                                                     | Review & Rating Module           |  | Suite 13: Reviews (`TC-REV-01..03`)     <b>**Module 14**</b>          |
|                                                                     | Reporting & Scam Flagging Module |  | Suite 14: Moderation Flags (`TC-REP-01..03`)     <b>**Module 15**</b> |
|                                                                     | Admin Management & Moderation    |  | Suite 15: Admin Controls (`TC-ADM-01..04`)     <b>**Module 16**</b>   |
|                                                                     | Notification Module              |  | Suite 16: System Alerts (`TC-NOTIF-01..03`)   +-----+-----            |

+-----+-----+ `` --- ## 6. Section E: Unclear  
Items & Clarifications 1. **\*\*Exclusion of Payment Processing:\*\*** Reconfirmed. The  
platform facilitates inquiries and in-person viewings; payments occur off-  
platform directly between parties. 2. **\*\*Dedicated `.drawio` Files:\*\*** 16 separate  
`.drawio` files will be placed inside `docs/flowcharts/`, giving the instructor a  
distinct diagram for every module.